

Domáca úloha č. 4

(4. L)

$$p(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0 \quad ; \quad a_0, a_1, a_2, a_3, a_4 \in \mathbb{R}$$

$$p(0) = 1 \quad ; \quad p(1) = 2, \quad p(2) = -3$$

$$a_0^2 + a_1^2 + a_2^2 + a_3^2 + a_4^2 \text{ hodnota najmenšia možná}$$

→ hľadáme riešenie sústavy s najmenšou normou

$$1) \quad 1 = a_0$$

$$2) \quad 2 = a_4 + a_3 + a_2 + a_1 + a_0$$

$$3) \quad -3 = 16a_4 + 8a_3 + 4a_2 + 2a_1 + a_0$$

$$\rightarrow \|a\|^2 = a_0^2 + a_1^2 + a_2^2 + a_3^2 + a_4^2$$

$$\rightarrow A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 \end{pmatrix} \quad b = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$$

$$\rightarrow a = A^T(AA^T)^{-1}b$$

$$(AA^T) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 5 & 31 \\ 1 & 31 & 341 \end{pmatrix}$$

$$(AA^T)^{-1} = \begin{pmatrix} \frac{486}{115} & -\frac{31}{46} & \frac{13}{230} \\ -\frac{31}{46} & \frac{17}{23} & -\frac{3}{46} \\ \frac{13}{230} & -\frac{3}{46} & \frac{1}{115} \end{pmatrix}$$

$$A^T(AA^T)^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{129}{230} & \frac{14}{23} & -\frac{11}{230} \\ -\frac{103}{230} & \frac{11}{23} & -\frac{4}{230} \\ -\frac{51}{230} & \frac{5}{23} & \frac{1}{230} \\ \frac{53}{230} & -\frac{4}{23} & \frac{17}{230} \end{pmatrix}$$

$$a = \begin{pmatrix} 1 \\ 4/5 \\ 3/5 \\ 1/5 \\ -3/5 \end{pmatrix}$$

→ polynom: $p(x) = -\frac{3}{5}x^4 + \frac{1}{5}x^3 + \frac{3}{5}x^2 + \frac{4}{5}x + 1$

$$\|a\| = \frac{2\sqrt{15}}{5} \approx 1,549$$